Tandem 16
Badge Building Instructions

Overview

The Tandem Badge project is a project that allows building and customization of a convention badge that can be worn at trade shows. The idea is a spin of the [DefCon badges](), and it leverages the [SAO]() interface as described in the [DefCon]() sites.

The badge is a 2-layer design that has the main electronics on the lower board with a top cosmetic PCB that covers the electronics. A paper white display provides a persistent name badge and is also the microcontroller for the electronics. The device is designed as a tribute to the original Tandem computer panel created in 1975-76 and used until the three production Tandem systems prior to the release of the VLX (T16 (aka TNS-I) TNS-II, and TXP). The panel on these is shown below. It has two LEDs that were used for displaying “power” and “run” (TNS-II and TXP only), and 16 more LEDs placed in a row to the right, next to the two other LEDs. The 16 LEDs were framed with a silkscreen to break the 16 bits into Octal values (groups of 3) with a single remaining LED to the left of the groups of 16. The silk numbers the bits in order from 0 (leftmost) to 15 (rightmost). Octal values are entered as big endian (e.g. %42 = 0 000 000 000 100 010).

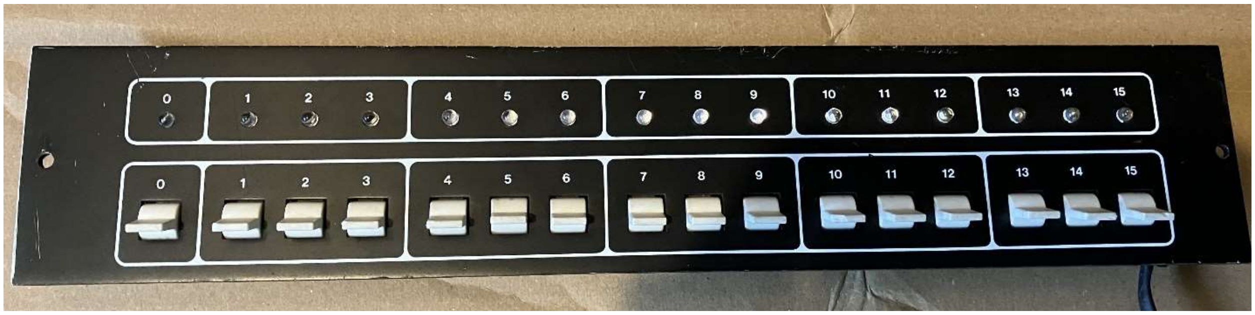


FIGURE 1- T-16 / TNS-1 SWITCH PANEL

Under the row of LEDs there are switches that set and reset the LEDs (in the real computer panel they also set/reset the registers in the computer).

For each switch, there is a corresponding LED above it. Individual switches can be set to any value.

The badge simulates the behavior of this design. It uses a “register” value to hold the current switch configuration. In some cases, the value of the switches causes special actions. These special switch values are documented elsewhere in the operation instructions on [Github]() (see above QR code).

The Tools required to build the badge

The electronics are placed using through-the-hole components. There are no surface-mount parts in this design. Soldering should be simple, even for beginning skills level soldering.

- Soldering iron and solder
- Ethyl cyanoacrylate “Crazy” glue (GELL stye greatly recommended)
- Small screwdriver (usually using a phillips screw)
- Pliers to hold the small nuts.

Parts List

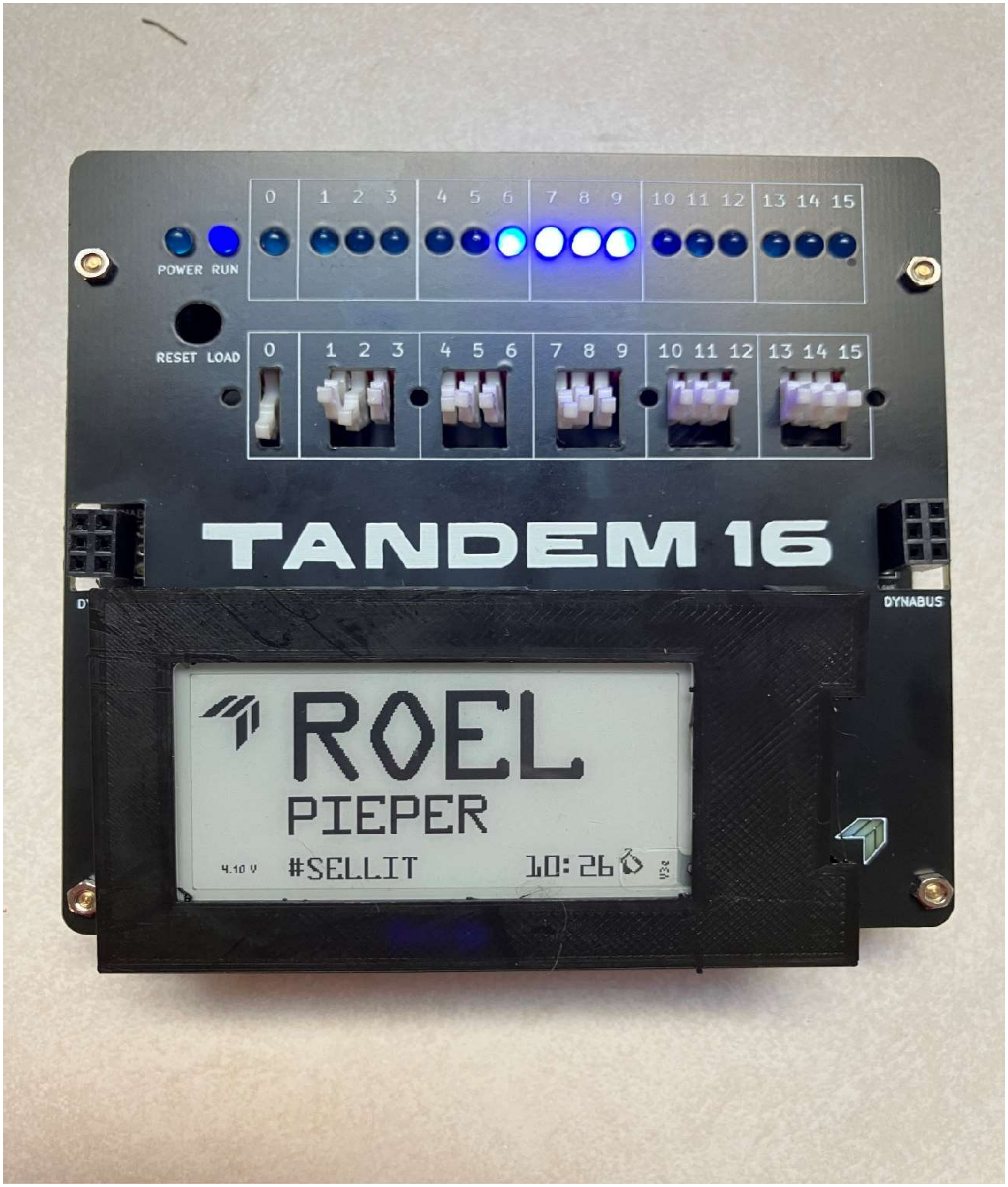
Badge parts

- 3D Printed parts (as part of the kit need to be printed prior to the build)
  - DIP switch extensions (16)
  - Display cover (optional shroud over the display to make it nicer)
  - Simulated reset key
- 3MM offset spacers 8-10MM long (4)
- 3MM offset spacers 3MM long (4)
- M2 Screws (4)
- M2 Nuts (4)
- Extended 2x3 female headers (2)
- 13 pin female headers (2)
- 13 pin extended (long) male headers (2)
- 9 pin resistor array (8 - 10k resistors) (2)
- 9 pin resistor array (8 - 470 resistors) (2)
- Small standard (3MM) LEDs (amber or yellow) (16) [Or blue for the Roel Pieper edition 😊]
- Small standard (3MM) LEDs (red) (2)
- Small standard (3MM) LED (green) (1)
- 3 switch DIP (MUST BE OFFSET, NOT STANDARD DIP SWITCH!) (5)
- 1 switch DIP (MUST BE OFFSET, NOT STANDARD DIP SWITCH!) (1)
- 67 ohm or near value (60-80 ohm) resistors (3)
- 74HC165 in DIP configuration (2)
- 74HC595 in DP configuration (2)
- DIP Sockets (16 pins) (4) (recommended but optional)
- Lithium battery 3.3V/4V
- Generic Diode (recommended) or a jumper.
- LilyGo ePaper (paperwhite) T3 display and ESP processor.
  - Software - Xinyuan-LilyGO/LilyGo-T5-Epaper-Series (github.com)

   - LILYGO® TTGO T5 V2.3.1_2.13 Inch E-Paper from Lilygo on Tindie (make sure includes the 9102 chip)

Optional SAO Parts

SAO boards use reverse gull-wing LEDs. Most also use SMD resistors of approximately 65-90 ohms. The header on each side is a shrouded male 6-pin. These should be the standard SAO compatible design. Each Tandem Badge SAO is different, but they all use the same header, resistors and LEDs.



Finished Build. This one was built with blue LEDs as a joke.

Step 0

Print or obtain the 3D printed items: 1 Key with base, 16 or more white lever extensions, and one display cover/shroud

Step 1

Place the longer 8-10MM stand-off spacers onto the main board from the top with the female facing down, Secure the shorter 3MM (or less) spacer using the male side through the hole to hold the top/longer spacer onto the main board. Repeat for each corner of the main board. In the end, you should have all 4 spacers placed with the short spacer on the bottom and the longer spacer on the top. This is a temporary placement of these stand-offs.

FIGURE 2- BOARD WITH SPACERS PLACED

Step 2

Place one of the three (~67 ohm) resistors into the resistor component holes. Orientation does not matter. Solder these leads to the board.

Place one of the red LEDs into the Throbber LED component holes. Note orientation. Anode (long leg) on the right, flat side left. DO NOT SOLDER the LED YET!

FIGURE 3 - LOWER RED LED PLACE LOOSELY

Step 3

Place the two leftmost top LEDs (red and green) into the D1 and D2 holes. MAKE SURE the ANODE is toward the lower side of the board and the cathode is toward the top edge of the board. Early versions of the board silkscreen have the A/K reversed. K should be on the top, and A on the bottom (with the flat side being the K at the TOP). This orientation is the same for ALL of the LEDS at the top of the badge.

Once the red and green LEDs are placed, place all of the remaining 16 yellow or orange LEDs into the remaining LED holes. Orientation should be the same for all LEDs. Flat side of the LED up toward the top of the board. Long lead toward the lower side of the board.

Step 4

Pick up the lower board and let the LEDs slide and rest onto the lower board (freely)   Same for the remaining red LED that sits on the lower right side of the main lower board.

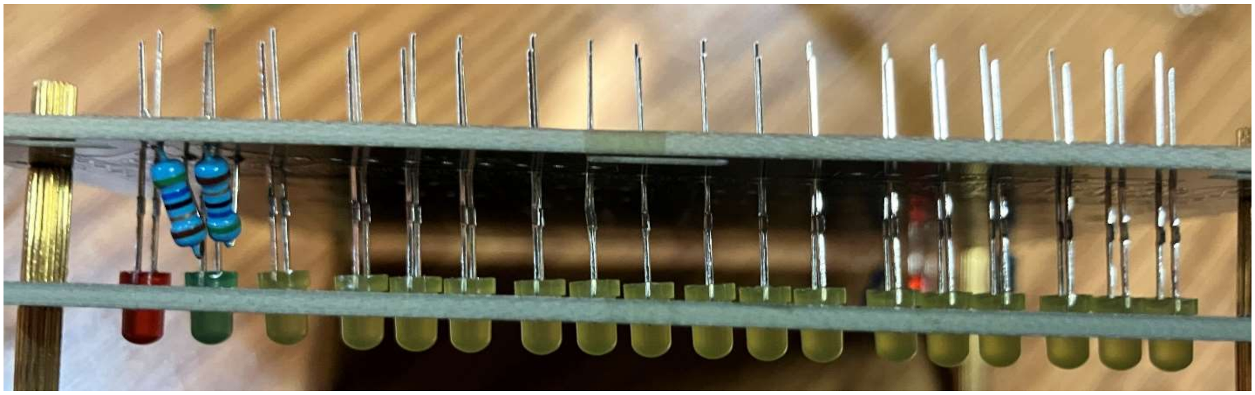
FIGURE 4 - LEDS PLACED LOOSELY INTO THE HOLES

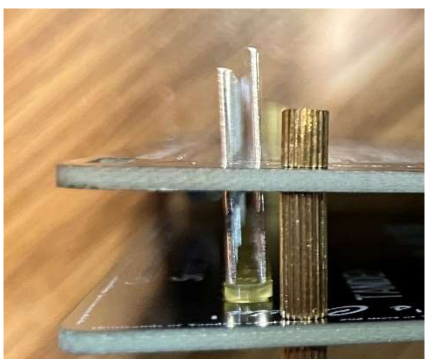
Rest the top board over the spacers of the bottom board. Install 2 to 4 screws to hold the top board onto the lower board. Then flip the entire badge over (both boards upside down). Guide each of the loose LEDs into the corresponding hole in the top board. Make each one fit directly into a hole and seat down all the way to the LED stopper. LEDs should all fit perfectly into each hole before proceeding. Check that ALL of the LED pins are aligned the same and that they are correctly placed before soldering.

Check the remaining red LED in the lower corner of the badge. Make sure it is resting with the tip on the little chevron on the top board. The LED should not be bent and should rest on the chevron before soldering.

Solder all LED connections now. Double and triple check orientation before clipping the remaining wire short.

Remove the top board. Leave the spacers on the corners. You will need those again on step 9.





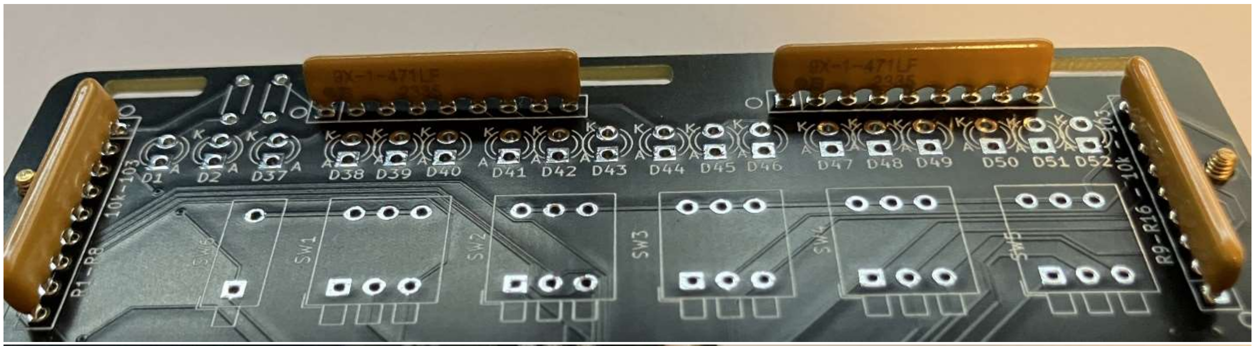
Step 5

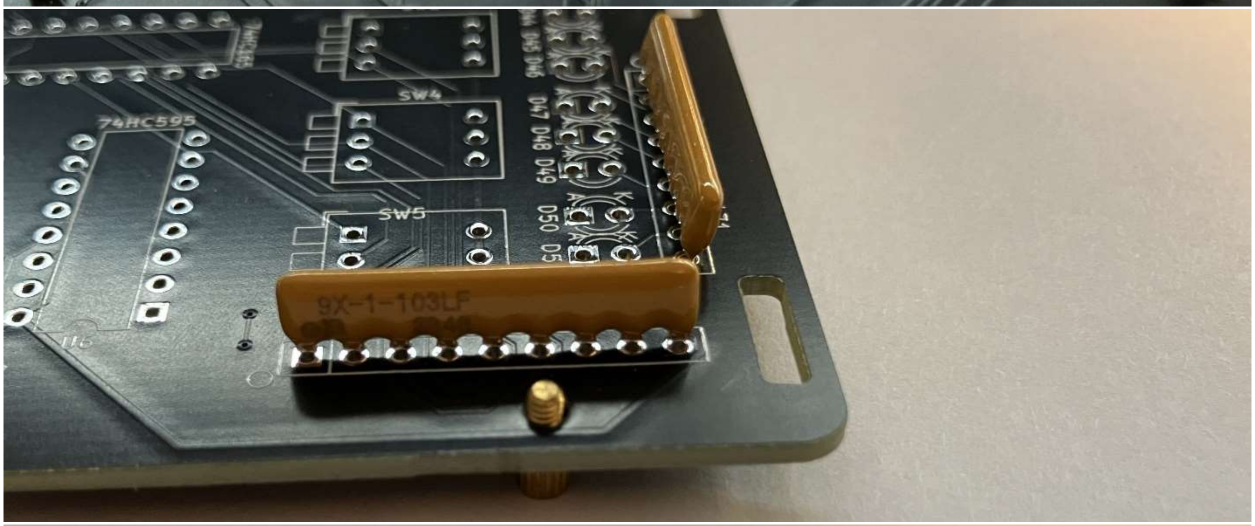
Place the diode or a jumper wire to the diode at the lower right side of the board. Make sure of the orientation. Cathode (bar on the diode) should be at the top hole. Solder and trip as appropriate.

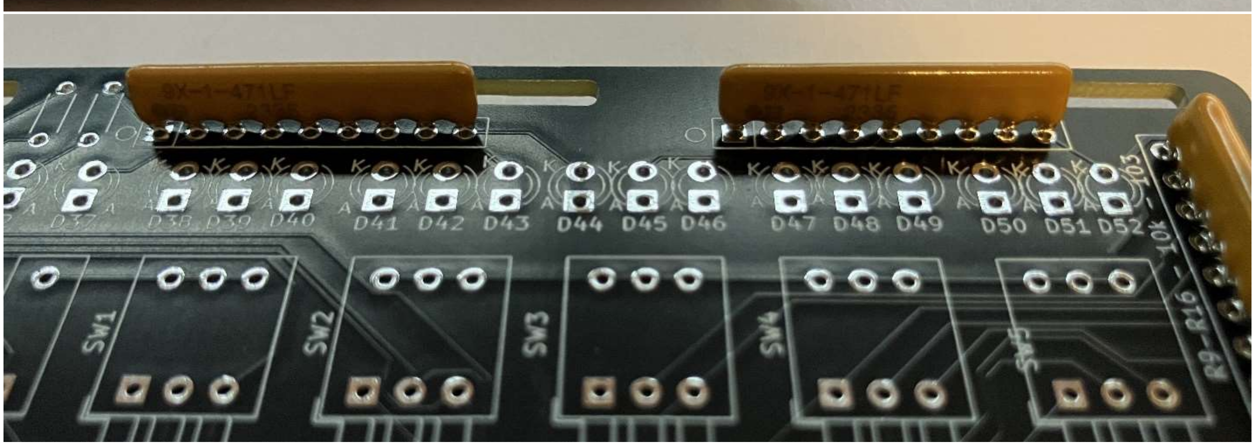
Step 6

Place resistor arrays. Be sure to place the proper value resistors into the correct locations and ensure that the orientation is correct. Match the dot on the board with the dot on the array.

The arrays at the top are marked 471. Align the dot on the array to align with the dot printed on the circuit board. (left of the board facing up). R17 through R34 are 471.





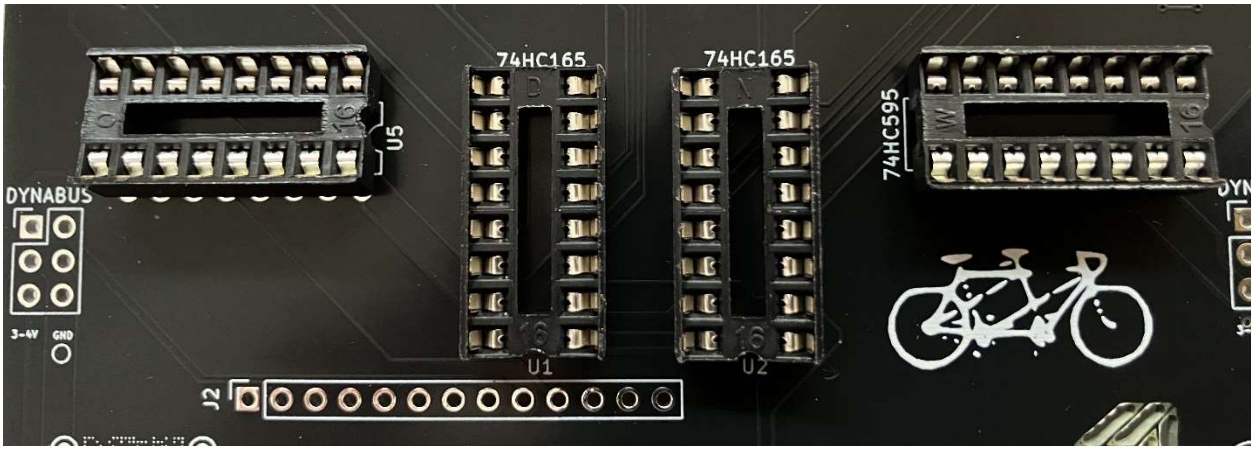


The remaining two 103 arrays are placed vertically into their locations. NOTE that these arrays are placed in opposing directions from each other. Make sure you place the dot with the dot on the array. When the 103's are placed, the leftmost should have the dot on top, while the rightmost one has the dot placed on the bottom.

Solder these to the board to complete this step.

Step 7

Place the four IC sockets. Orientation of these sockets is shown on the board. Do not mount the ICs into the socket yet. Solder all of the connections to the board. If you make a mistake on orientation, it is not a disaster. But make sure you note it when you place the ICs. The sockets can be wrong, but the IC insertion (later) cannot be.



Step 8

Place the DIP switches. 5 sets of three, and one set of one. The easiest placement is to do one at a time from one side to the other. Or, you can place them all and solder them all.   Caution to be sure that the DIP switch is FACING TOWARD THE BOTTOM. If you invert these, this will be really annoying to fix. (Not that this has ever happened before?)

Step 9

Place all of the headers.

This step is best done with the top board placed temporarily. Place the top board back onto the spacers and secure it to the stand-off using another stand-off spacer that is as deep as the 6-pin female header with the long pins. In other words, place the stand-off so that it can be used to space the header to the table when the device is upside down.

If this does not make sense, the objective is to solder the two female headers to the “dynabus” circuit board holes such that the most possible length is used. There should be just enough length on the headers to be soldered onto the lower board. Do this for both 6-pin headers. The header should protrude fully through the top board. DO NOT PLACE IT LEVEL WITH THE TOP BOARD. IT NEEDS TO PROTRUDE above the top board. See the diagram if this is not clear.

FIGURE 5 - BOTTOM OF BOARD SHOWING THE FULLY EXTENDED LONG PINS

FIGURE 6 - FEMALE HEADERS EXTENDED THROUGH THE TOP BOARD.

Break the female header lengths into two 13-pin lengths.   Plug the 13-pins into the LONG male header pins. DO not use standard length male headers. Use extended length ones. Two 13 pin headers should be placed through the top board and into the bottom holes. Before soldering anything, place the display board onto the male pins that protrude through the entire kit. Make sure all pins are aligned on the display card. Solder the display to the male header pins first. The heat dissipates and the soldering is more difficult than on the PCB. This is normal. Use flux if you need to. Be patient and let the component warm enough to deliver the solder for each of the 26 solder points on the display. Be careful, but persistent with each solder taking a long time because of the heat dissipation. It is normal for this to be a tie-consuming step. A good connection in all pins is critical here.

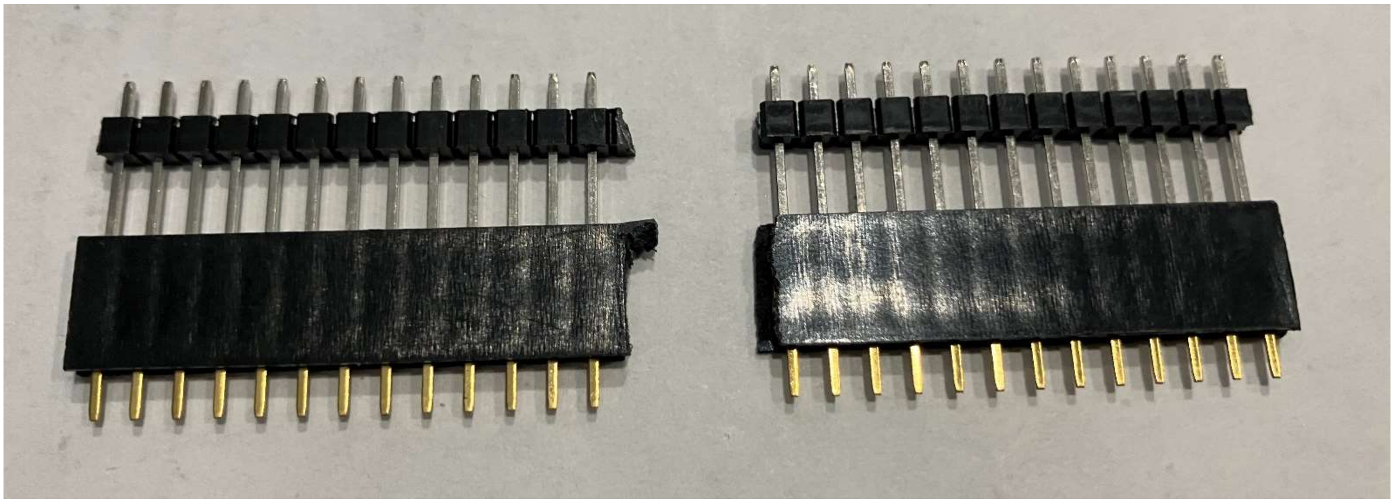


FIGURE 7 - HEADER PINS TRIMMED AND PREPARED FOR PLACEMENT

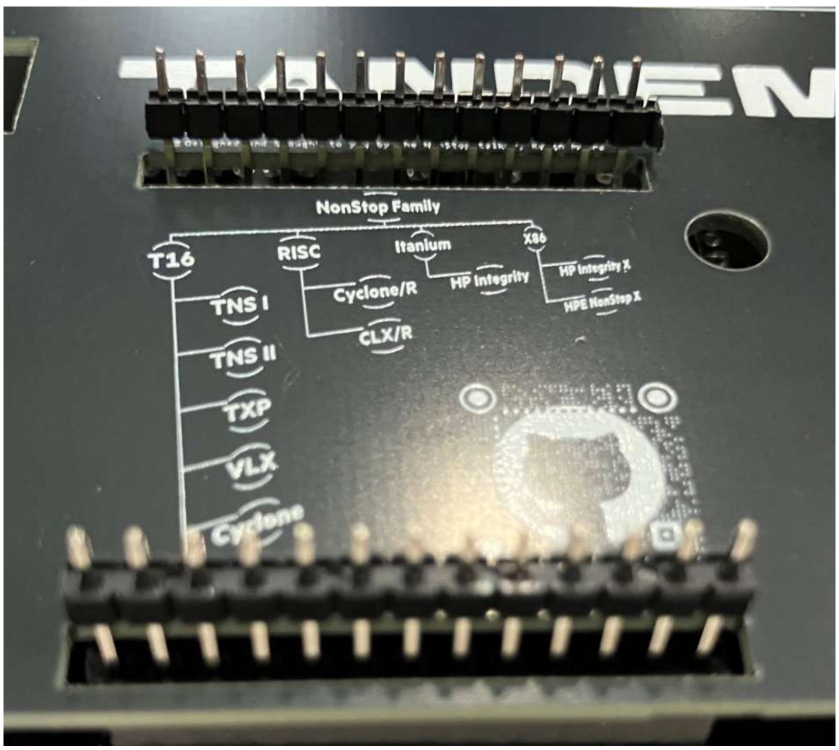


FIGURE 8 - HEADERS PLACED PRIOR TO SOLDERING -NO SOLDER ON BOTTOM SIDE YET

FIGURE 9 - DRY PLACEMENT OF THE DISPLAY ON TOP. SOLDER THESE PINS FIRST

FIGURE 10 - SOLDER BOTTOM PINS AFTER THE DISPLAY-SIDE PINS ARE SOLDERED

Once the display header is completely soldered. Carefully flip the kit over and solder the two 13 (26) pins into the board on the bottom.

Step 10

FIGURE 11 - DISPLAY AND TOP BOARD REMOVED. THE FEMALE HEADERS SHOULD BE SOLID NOW.

Place the components onto the main board and prepare the battery cable.

Unplug the display from the female header. Set aside. Remove the top board again. Place the integrated circuits into the sockets and make sure the orientation is correct to the marking on the board. Make sure that the proper chips are placed into the proper socket locations on the board. The 595 are horizontal and the 165 are vertical.

Step 11

Final assembly.

If you have not placed the little miniature key, it can be placed now. The key goes through the top hole and the little cap captures through the hole and locks with a gentle 1/8 turn. Be very gentle. It is easy to break the stem. It only has to turn just enough to set. Less than 1/8 of a turn should suffice. If required you can fabricate some tape below (or glue) this cosmetic part. When it works, it will spin nicely. But it is not a necessary part at all.

Look over everything If everything is correct and soldered, process to place the battery between the female headers with the wire protruding to the right side. Place the wire through the small circular

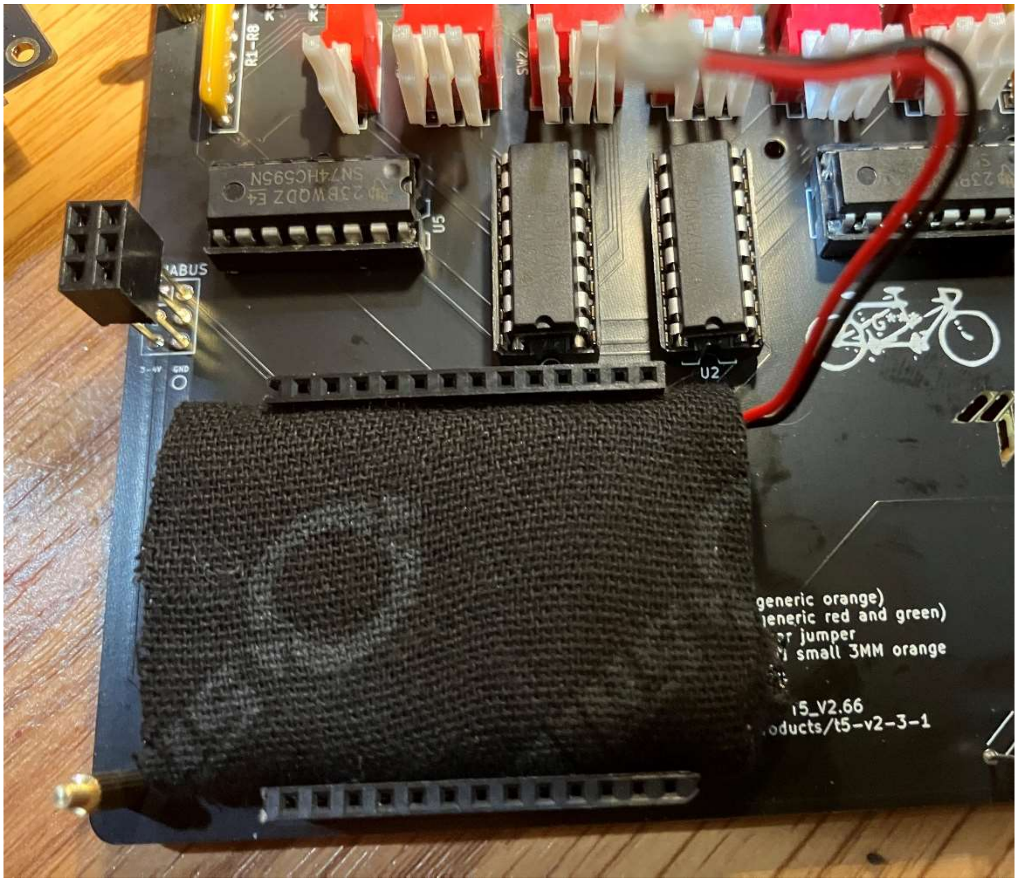


FIGURE 12 - BATTERY MAY APPEAR DIFFERENT

hole on the top board.   Secure the top board to the stand-off spacers with screws or nuts depending on which way the header is facing.

If you have a shroud for the display, place it over the display and touch the plastic pins that go through the display corner holes with the soldering iron to melt the 4 corners in place.

Once the top is secure, first, plug the battery into the female plug end center bottom of the display board, then carefully insert the display header back into the female header through the slot in the top board (like you did when you soldered it in place). The battery plug is keyed and can only go in one way. Do not force it the wrong way.

Optionally (recommended) place a dummy bottom board with the short stand-off spacer. The dummy board is just used to prevent catching the bottom pins against anything – and prevents short circuits. The dump board can be oriented in any way or can be either a top or bottom dummy board. I use old bad boards. One should be provided.

Step 12

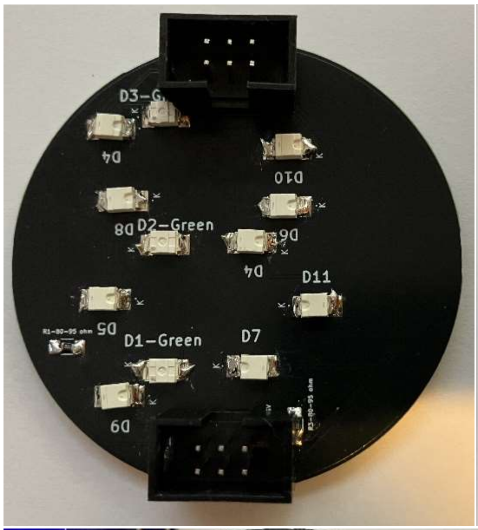
Carefully turn on the device by sliding the little switch on the display. Left is on, right is off. The two push switches are for reset and for programming via interrupt. The reset can be used to reboot. The other button is not used.

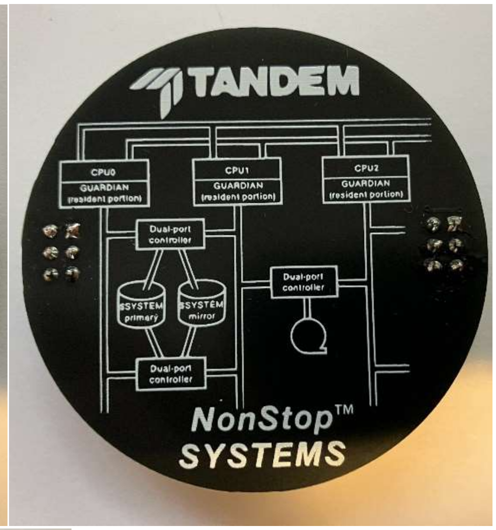
If the unit is programmed, you can proceed to configure the memory for personalization. If not, the processor will need to be programmed using Arduino or AVRdude. Instructions for that are elsewhere.

The SAO board(s) (Optional)

The SAO boards use reverse gullwing LEDs (11) and small SMD resistors(3). Any can be used and the board shows the location and range of possible components. LED orientation is marked on the board with a dot for the cathode (K).

Use Male header with shroud for the pin connector to the main board. The SAOs designed for this project will have pins that can be placed onto either side of the main board “dynabus” header.





Final Step

Enjoy!

I am at nonstoptalker@hpe.com or keithmoore@acm.org.